Superstore Sales Analysis Using Power BI

Project Documentation

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1. Introduction

Businesses collect sales data every day, but raw data is not always easy to understand or use. This project uses Microsoft Power BI to explore a Superstore sales dataset and turn it into useful information through charts, key performance indicators (KPIs), and interactive dashboards.

1.1 Problem Statement

It can be difficult for a business to identify sales trends, compare product categories, understand regional performance, and review customer segments by looking at raw tables alone. The project aims to present these details in a clear visual report that supports better business decisions.

1.2 Project Aim

The main aim is to analyze the available Superstore sales data and build an interactive Power BI dashboard that summarizes sales performance and highlights useful patterns.

2. Project Planning and Management

2.1 Project Objectives

- Understand and prepare the dataset before analysis.
- Analyze total sales and changes over time.
- Compare sales across product categories and sub-categories.
- Explore sales across regions, states, and cities.
- Compare sales across customer segments and shipping modes.
- Create interactive Power BI pages with clear charts, filters, and KPIs.
- Present findings and practical recommendations based on the results.

2.2 Project Scope

Included in the Project	Not Included in the Current Scope
Data inspection and cleaning	Machine-learning forecasting
Sales KPI calculations	Live connection to a sales system
Time-based sales analysis	Inventory management
Product, customer-segment, and geographic analysis	Profit analysis if profit data is unavailable
Interactive Power BI dashboard	Discount and quantity analysis if those fields are unavailable

The analysis will be limited to the fields available in the supplied file. Before implementation, the team will confirm the dataset schema. If fields such as Profit, Quantity, or Discount are not present, the project will not claim to calculate or analyze them.

2.3 Project Timeline

Week	Tasks	Expected Output
Week 1	Define the problem, goals, scope, and review the dataset.	Project plan and data overview
Week 2	Clean and transform the data using Power Query.	Prepared dataset
Week 3	Create DAX measures and build dashboard visuals.	Initial Power BI dashboard
Week 4	Test calculations, summarize findings, and prepare the report and presentation.	Final dashboard and documentation

2.4 Team Roles and Responsibilities

Role	Responsibilities
Data Cleaning and Preparation (zaid yousef)	Inspect the source dataset; identify missing values, duplicates, inconsistent labels, and incorrect data types; clean and transform columns in Power Query; document changes and validate the prepared data.
DAX Measures and Calculations (yousra mohamed) (kareen badrouse)	Create and test DAX measures after cleaning, including Total Sales, Total Orders (distinct Order ID), Total Customers (distinct Customer ID), and Average Order Value; validate results against the source data.
Data Model Design (ahmad sameh)	Build the Power BI data model; define table relationships and cardinality; configure date relationships where applicable; check filter direction and avoid ambiguous relationships.
Visuals and Dashboard Design (mohamed hisham) (Mahmoud yaser)	Create report pages, KPI cards, charts, slicers, and useful tooltips; keep formatting consistent and labels clear; verify visuals respond correctly to filters and reflect validated measures.

The team should follow this workflow in order: (1) clean and prepare the data, (2) create and validate DAX measures, (3) build and verify the data model, and (4) design and test the visuals. Roles can be combined or reassigned depending on the number of students in the team.

2.5 Potential Risks and Mitigation

Risk	Possible Effect	Mitigation
Missing or inconsistent values	Incorrect or incomplete results	Inspect columns and handle missing values carefully.
Incorrect data types	Date and numerical calculations may fail	Set and validate the correct data types in Power Query.
Incorrect DAX measures	Misleading KPIs	Check measures against simple calculations from the source data.
Crowded dashboard	Users may find results difficult to understand	Use a consistent layout and focus on relevant visuals.
Limited dataset fields	Some business questions cannot be answered	Document data limitations and avoid unsupported conclusions.
Time delays	Incomplete project deliverables	Divide work into weekly tasks and review progress.

2.6 Success Measures

The project will be considered successful if:

- The dataset is checked and prepared correctly.
- The main measures agree with the underlying data.
- The dashboard filters work as expected.
- The report clearly presents sales patterns by time, product, geography, and customer segment.
- The team provides conclusions that are supported by the data.
- The final report and presentation are submitted on time.

3. Literature Review

A literature review helps the team understand existing business-intelligence tools and common approaches to sales analysis. For this student project, the review focuses on dashboard reporting, retail sales analysis, and a structured data-analysis process.

3.1 Existing Tools and Approaches

Microsoft Power BI

Power BI is a business-intelligence tool used to connect to data, transform it, create visual reports, and share interactive dashboards. It supports charts, slicers, filters, and drill-down features. For this project, it will be used to present sales information in one report.

Retail Sales Analysis

Retail sales analysis commonly compares performance over time, across product groups, customer segments, and geographic areas. These comparisons can help identify strong and weak areas of sales. The usefulness of the analysis depends on the quality and coverage of the available data.

CRISP-DM Methodology

CRISP-DM is a structured approach to data-mining projects. Its main stages include business understanding, data understanding, data preparation, modeling, evaluation, and deployment. This project can use the relevant stages as a guide, with the final deployment stage represented by the completed Power BI report and its presentation.

3.2 Comparison with the Proposed Project

Criteria	Common Existing Approach	Proposed Project
Data source	Depends on the business or sample dataset	Supplied Superstore sales file
Presentation	Static reports or interactive dashboards	Interactive Power BI dashboard
Time analysis	Sales trends by day, month, or year when dates exist	Sales trends using the available order dates
Product analysis	Category or product comparisons	Category, sub-category, and product comparisons
Geographic analysis	Regional comparisons when location fields exist	Region, state, and city comparisons
Customer analysis	Customer groups when segment data exists	Sales by the available customer segments

3.3 Project Contribution

The project will bring several sales views together in one interactive report. Users will be able to filter results and compare time periods, product groups, locations, and customer segments. The goal is not to create a new analysis method, but to apply established data-analysis and visualization practices to the supplied dataset and communicate the results clearly.

4. Requirements Gathering

Requirements gathering identifies who will use the report, what information they need, and which features the Power BI dashboard should provide. Requirements will be checked against the columns available in the dataset.

4.1 Stakeholders and Their Needs

Stakeholder	Main Need
Sales Manager	Review total sales and changes over time.
Business Owner	Compare products and regions to understand sales performance.
Marketing Team	Understand sales distribution across customer segments.
Operations Team	Review shipping modes and shipping duration where dates are valid.
Data Analyst	Explore data, validate measures, and identify patterns.
Project Supervisor	Review the method, results, and quality of the final project.

4.2 Functional Requirements

ID	Requirement
FR-01	The system shall import the supplied dataset into Power BI.
FR-02	The team shall inspect data quality and clean or transform fields where necessary.
FR-03	The dashboard shall show total sales and other valid measures supported by the dataset.
FR-04	The report shall show sales trends over time using the available date fields.
FR-05	The report shall compare sales by category, sub-category, and product.
FR-06	The report shall compare sales by region, state, and city where location values are valid.
FR-07	The report shall analyze sales by customer segment when segment data is available.
FR-08	Users shall be able to filter report visuals using relevant slicers.
FR-09	The final report shall communicate findings and data-supported recommendations.

4.3 Non-Functional Requirements

Requirement	Description
Accuracy	Measures and visuals should be checked against the source data.
Performance	Visuals and filters should respond smoothly for the size of the dataset.
Usability	The dashboard should have clear labels, readable charts, and a consistent layout.
Security	The team should avoid exposing personal or sensitive information unnecessarily.
Maintainability	Measures and fields should have clear, consistent names.
Reliability	Filters, visuals, and calculations should behave consistently.
Compatibility	The report should work in Power BI Desktop; sharing through Power BI Service depends on access and licensing.

4.4 Data Requirements

Field or Field Group	Purpose
Order Date and Ship Date	Time analysis and shipping-duration calculations.
Order ID	Count distinct orders rather than counting data rows.
Customer ID and Segment	Distinct customer counts and customer-segment analysis.
Category, Sub-Category, Product Name	Product and category comparisons.
Sales	Total sales and sales comparisons.
Region, State, City	Geographic analysis.
Ship Mode	Compare orders by shipping method.

The team must verify the actual column names and data types before implementation. A single order may appear in multiple rows because it can contain multiple products; repeated Order ID values do not automatically mean duplicate records. Profit, quantity, and discount analysis should only be included if the corresponding fields exist in the supplied file.

5. Proposed Dashboard Design

Page	Suggested Content
1. Sales Overview	KPI cards for total sales, distinct orders, distinct customers, and average order value; monthly sales trend; date and region slicers.
2. Product Analysis	Sales by category and sub-category; top products by sales; category contribution.
3. Geographic Analysis	Sales by region, state, and city using bar charts or a map where appropriate.
4. Customer and Shipping Analysis	Sales by customer segment, orders by ship mode, and shipping duration if the dates support it.

6. Initial DAX Measures

The following are example measures. The table and column names must be adjusted to match the imported dataset. These measures should be validated after data preparation.

Total Sales

```
Total Sales = SUM('Superstore'[Sales])
```

Total Orders

```
Total Orders = DISTINCTCOUNT('Superstore'[Order ID])
```

Total Customers

```
Total Customers = DISTINCTCOUNT('Superstore'[Customer ID])
```

Average Order Value

```
Average Order Value = DIVIDE([Total Sales], [Total Orders])
```

7. Expected Outcomes

- A cleaned and prepared dataset suitable for analysis.
- A Power BI report with interactive pages and filters.
- A set of validated sales KPIs and visual comparisons.
- A summary of important patterns and findings supported by the data.
- Practical recommendations based on the results, with limitations clearly stated.

8. Conclusion

This project applies data-analysis concepts and Power BI visualization to a Superstore sales dataset. By preparing the data, defining clear requirements, and designing an interactive dashboard, the team aims to make sales information easier to explore and understand. The final conclusions will be based on the actual results produced during analysis, and any limitations in the dataset will be stated clearly.